

Architecture: Mini-moi

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Maintained baseline, reviewed through 2026-07-21. This document separates the design principles that have held since inception from the current implementation. It identifies where real growth has caused drift and ties each gap to follow-up work.

What This Is

Mini-moi is a personal AI agent platform made up of purpose-built domains. Each domain addresses a different part of daily life while following the same basic pattern: take in signals, use models suited to the task, build a profile of what matters, produce something personalized, and learn from the response. The domain content changes. The architecture remains consistent.

The deeper design goal is longitudinal. The platform does more than process content; it captures *thinking about* content. That includes reactions, curiosities, areas marked for deeper investigation, and responses to what the system itself notices. Those signals become a growing record. Nine to twelve months from now, the accumulated asset should not be an archive of articles or transcripts. It should be an evolving map of what one person thought, what they returned to, and how their emphasis shifted as the world changed. This is the standard against which the architecture should be reviewed: does the design still serve that goal?

The pattern also applies beyond these domains. Input, analysis, a persistent and updating profile, personalized output, a closed feedback loop, and a thinking record that compounds are the same elements an organization would need to build an AI system that improves at its job over time. The domain content is interchangeable. The architecture is the durable asset.

The platform began in 2025 as a one-day experiment. A small Python and LLM system used two personal data sources, email and bank statements, to produce an expense management dashboard. It ran entirely on a local Ollama model. The local design served two purposes: protecting private financial data and testing what local models could actually do. Both reasons still apply. Local capability is also likely to matter more as open models and local hardware improve, including in physical AI. The platform grew from that experiment without abandoning its original privacy and independence commitments.

The system has been in daily production use since February 2026 and currently runs five domains.

The Five Domains

Curator. Curator is the oldest domain and the one with the longest record of daily production use. Each morning it pulls roughly 700 candidates from international RSS feeds, enriched X bookmarks, and targeted web search. The topics are focused, but the sources are deliberately broad. An untargeted serendipity pool adds

"friction" by surfacing material the learned profile might otherwise suppress. This reduces the risk that previous choices turn into a self-reinforcing filter.

The top 20 articles arrive each morning, but delivery is only the start of the loop. Reactions, saved articles, Deep Dive requests with stated motivations, and responses to the system's own observations are also captured. Together they form a record of reading and thinking over time. The mobile site is the primary interface. The laptop site supports more comfortable reading and commentary. Telegram, the original pre-UI channel, remains available for notifications and as an alternate path.

Mein Deutsch. The German domain is built around simulated immersion rather than a course or coaching program. Unlike real-world immersion, its practice can be repeated and modified freely. It addresses three practical gaps: too few opportunities to speak; authentic reading that can become burdensome without well-placed support; and too few reasons to write something interesting in context and receive useful correction. Live voice conversation with AI personas (Gespräche) remains the central feature. Reading with contextual hints and translation (Lesen), writing with correction (Schreiben), vocabulary drills (Wörter), and the session archive (Archiv) reinforce one another around it. Human tutoring is supported alongside the AI practice. The domain also proved useful in real travel: difficult interactions could be recreated later as practice scenarios and repeated before the next encounter.

Meu Português. Portuguese was built as an intentional extension of the language architecture, with a third language already in view. Family members brought varied levels of fluency, exposure, and formal practice, along with a shared desire to maintain and develop their existing abilities. Each authenticated user has separate data, progress, and custom conversation personas.

The first release deliberately emphasized the frontend, multi-user separation, and low-friction access. Features that require more setup or subscriptions, such as Anki generation and some mobile voice paths, were outside that initial scope. The implementation moved quickly so it could enter real family use and extend the broader mini-moi pattern beyond one language. The remaining differences from German reflect staged validation and intentionally different scope, not a failure to follow the German design. Convergence is still important before a third language is added.

Guild. Guild is the platform's workshop. Its three areas are **Build • Operate • Improve**, and its working cycle is *spec → build → operate → improve*. Specifications become build items, status remains visible, and operations are controlled here. This shared view helps the operator follow continuous planning and development, a need that becomes more important as multiple coding agents contribute. Guild's scope is expanding toward stronger operations control and a more explicit learning function. Differences revealed by staged builds, along with lessons from operation and new techniques, feed the next specification. The planned **Master Craftsman** will coordinate build quality and standards inside Guild, with local access and a role kept separate from CoS.

Chief of Staff (CoS). CoS is the newest domain. It was separated from Guild once their different purposes became clear. CoS is designed as a working partner, not an executive assistant or secretary. It should retain the reasoning behind decisions, not only the task list; act within defined limits; watch across domains; surface unrequested but relevant information; and disagree when the evidence points away from the easy answer. Its current functions include a daily executive briefing, conversation, note-taking, and decision and action logs. Carrying intent with bounded authority is the longer-term role, and that work is still at an early stage.

All five domains emerged from actual use. Each exists because it met a real need, and daily usefulness continues to drive focus and priority. This creates a deliberate rhythm. Some phases move quickly because use demands a capability; others step back to rationalize what was built. The drift documented below is a visible result of that rhythm, and the follow-up work is how it is corrected.

Current State at a Glance

Domain(s)	Maturity	Key current gaps
Curator	Mature; daily production since Feb 2026	Model override ignored (intended fallback still used); Deep Dive script consolidation; AI Observations automatic daily and weekly runs awaiting production verification
German / Portuguese	Active twins; identical from the user's perspective, converging underneath	Bidirectional normalization before French; translation model choice not yet backend-configured; production Lesen refresh safe but not yet scheduled
Guild / CoS	Recently split; both in active redesign	CoS partner layer just starting (Python-only today, bounded OpenClaw shortly); accumulate-then-synthesize pattern not yet extended here

Design Principles

Most of these principles were established in the March 2026 production document, before the platform reached its current scope. Real growth has tested them since. The final two were added after that growth exposed new requirements.

1. Local-first data. Learned state lives in application-owned files. The files are structured so they can migrate to a relational store when volume or query complexity justifies the additional infrastructure. Preferences and history remain with the person rather than any one model, provider, or machine.

2. Model-agnostic and multi-model. Personalization is injected at the dispatcher level, above the individual model call. A model can therefore change without rewriting what makes the output personal. This is an operating discipline with three practical benefits.

Cost: model choice is the platform's main cost lever. A model with better capability at lower cost can be added, evaluated against the current option, and adopted if it performs better. This add-evaluate-switch cycle is normal operation.

Capability: different jobs need different strengths. Fluent spoken German and bulk article filtering are not the same problem, and no provider is best at every task.

Independence: the system is designed to retain a fully local path. It ran on local Ollama before any cloud model was used and should fall back to local operation at zero marginal cost if cloud providers become unavailable. As open models and local hardware improve, local inference is expected to handle more of the platform's work, including coding. Some call sites remain hard-coded; those gaps are documented below.

`tests/test_local_llm.py` is part of the standard regression suite. On the development host, where a local model server is available, it makes a real call to the configured local model whenever the suite runs. The test protects end-to-end execution of the local inference path rather than measuring search or task quality. GitHub-hosted runners skip only this live call because they do not provide the machine-local model service. This keeps the zero-marginal-cost path exercised continuously and prevents it from silently degrading while cloud models handle production work.

3. Swappable architecture, not merely swappable configuration. Components should be replaceable without breaking the rest of the system. This is a structural goal, not a claim that every call site has already reached it. Principle 7 and the current-state notes identify where implementation still falls short.

4. Quiet paths over noise. If there is nothing meaningful to say, nothing is sent. No domain pads its output to appear active. This applies to Telegram, logs, and the rest of the platform.

5. The operator stays in control. The build workflow follows a fixed path: design, human review, build, human confirmation, and ship. It does not currently allow autonomous agent-to-agent execution. A July security remediation followed this process while finding and closing real cross-user data exposure risks.

The principle can evolve without weakening. The planned next step is bounded agent-to-agent handoff, starting with CoS coordinating with Guild's Master Craftsman agent. CoS should notify the operator and gather information across agents, while the operator remains the final decision point. The goal is a single point of contact, not reduced control.

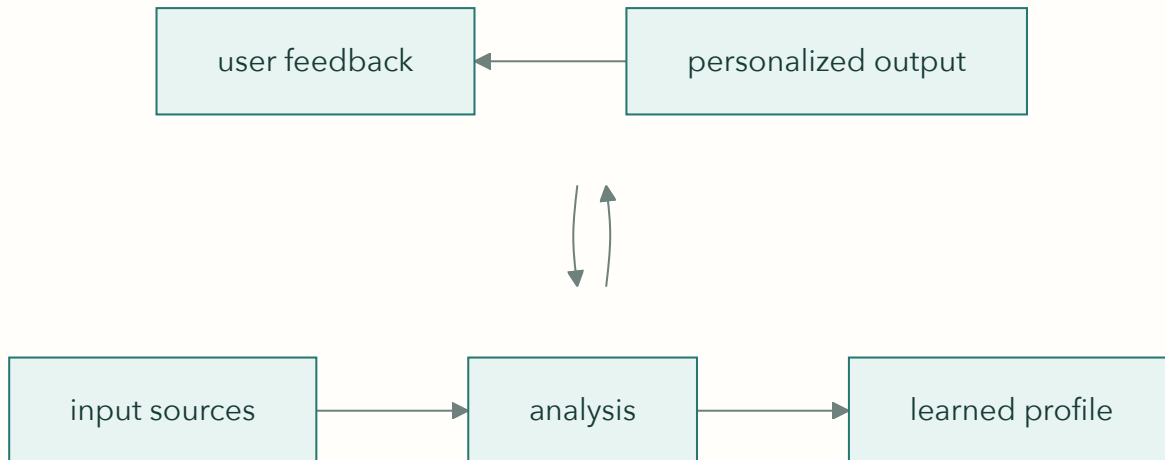
6. Extract shared code only after duplication has proved the pattern. A new domain may begin by copying an existing implementation. Shared code should be extracted only after two domains have demonstrated the same approach and started to converge. Abstracting earlier is premature.

Lesen and Leitura are a concrete example. Their reading flows reused proven Curator patterns by copying and adapting them for language practice. Curator was not turned into a general-purpose, multi-domain reading component. Shared code becomes appropriate only after the German and Portuguese implementations have both proved the pattern and begun to converge.

7. Verify production reality. Documented intent and running behavior are different claims. Recent checks found a directory that appeared committed but was not, a volume mount with no sync step, an incomplete backup script, inconsistent identity labels, and a configured model override that was ignored while the intended fallback model still ran. Every case looked correct in code or documentation. Outputs and logs exposed the difference. Production behavior must be verified, not inferred.

The Domain-Agnostic Pattern

Every domain is an instance of the same loop. The March 2026 platform document described this pattern before German, Portuguese, or CoS existed:



Curator: RSS and X bookmarks → AI scoring → learned preferences → daily briefing → reactions, Deep Dive requests, and observation responses.

German and Portuguese: sessions and transcripts → correction and analysis → per-user progress → next practice material → corrections accepted or revised.

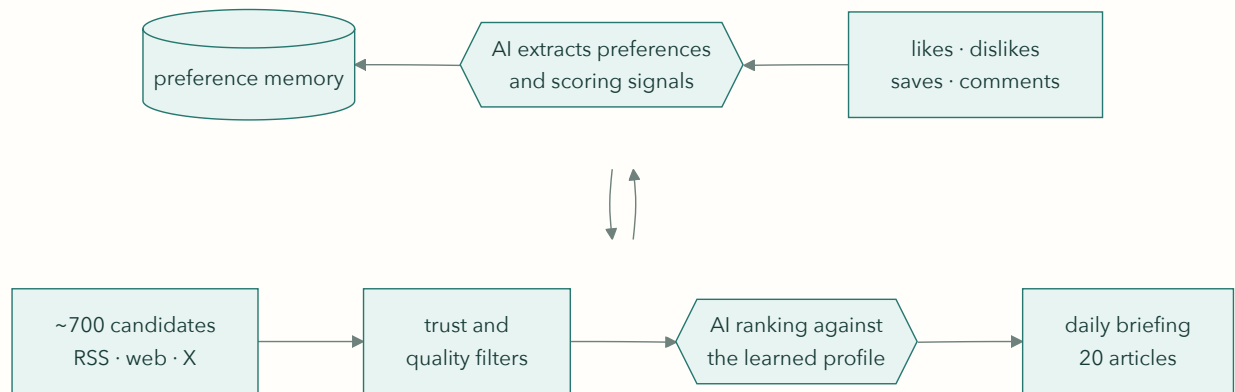
Guild and CoS: specifications, build status, and external signals → triage and synthesis → build queue and decision log → briefings and recommendations → approved, revised, or rejected.

The same shape appears five times. The full accumulate-and-synthesize loop is proven end to end in Curator (see Learning Loop below). Repetition across five different domains is the evidence for "platform over product."

AI Usage Across the Platform

Two capabilities are the clearest current examples of AI use: Curator's path from daily reading into deeper research, and live voice conversation in the language domains. Guild and CoS serve broader platform operation and coordination and are less mature as AI loops. Each domain uses AI differently because the model architecture is fitted to the job, not imposed uniformly. Multi-level model use appears in both Curator and the language domains, but for different reasons.

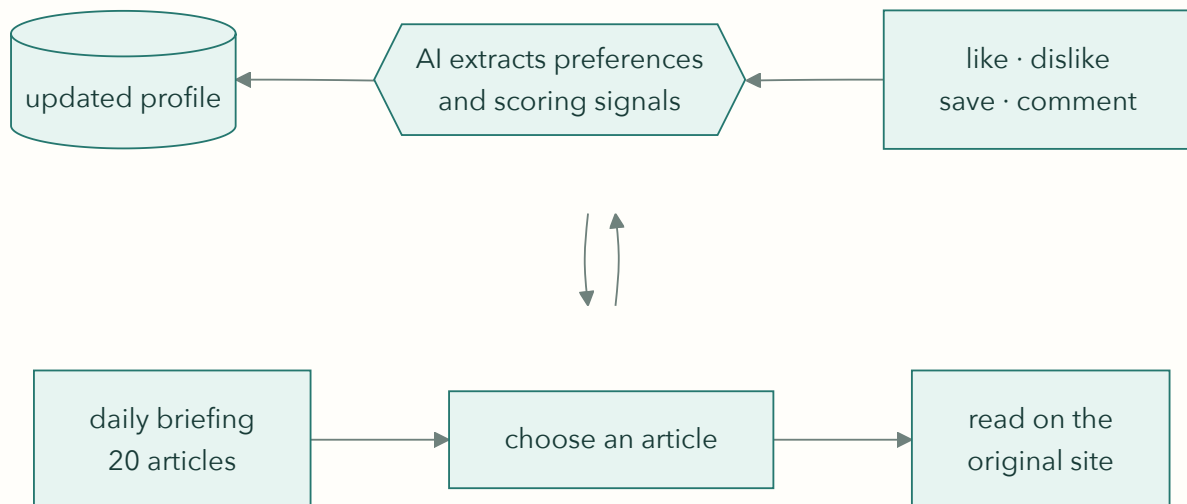
Curator: the mature, flagship pattern



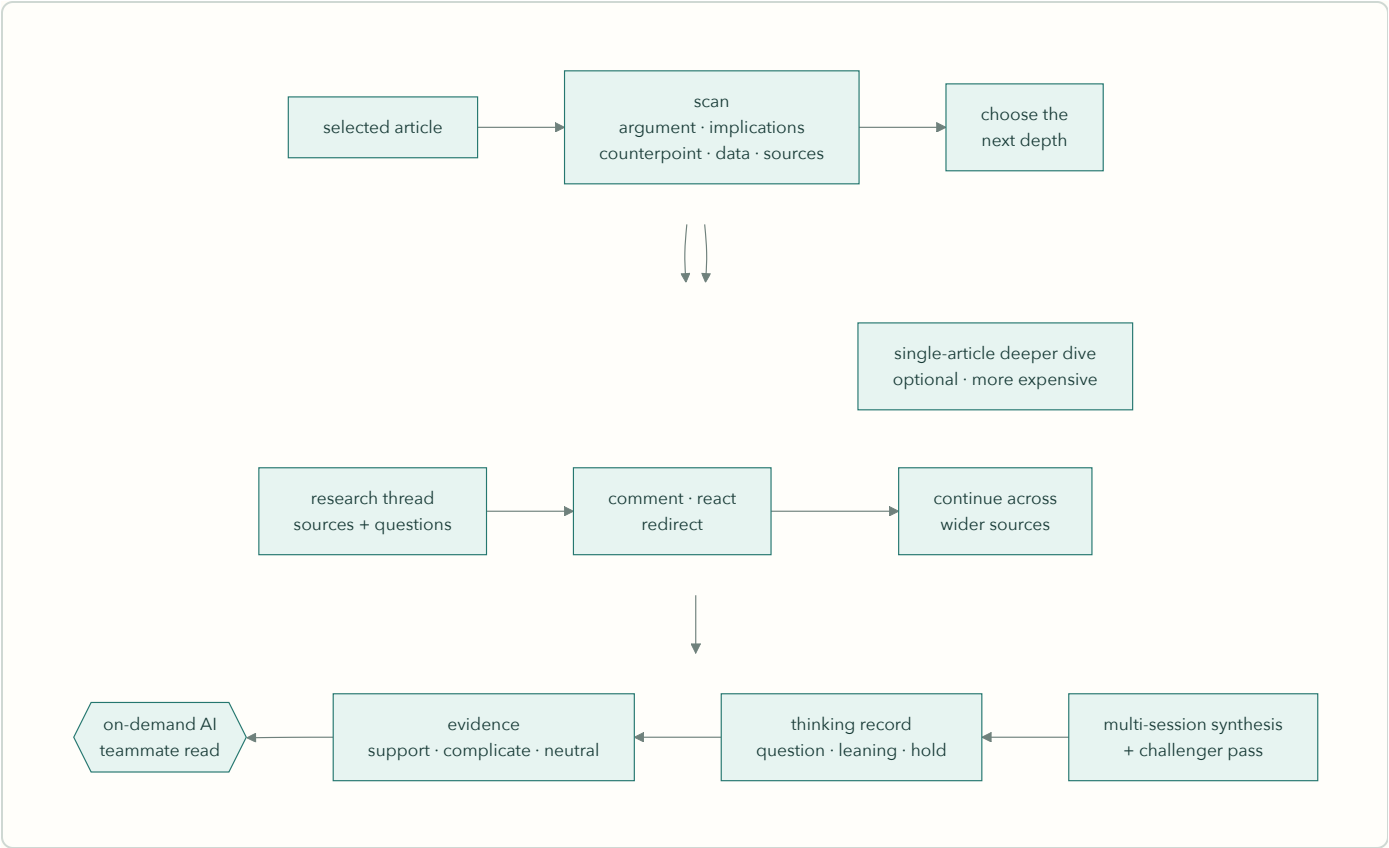
The production ranking path is deliberately single-stage today: one model pass over the full filtered pool. A two-stage mode, using a lower-cost pre-filter followed by a higher-capability ranking model, is available per run but is not the production default.

The closed selection loop is implemented: article feedback updates learned patterns, and those patterns are injected into later scoring. From the daily briefing, Curator also opens a longer research path:

Reading and feedback



Research and the thinking record



Not every stored response closes an automatic loop yet:

Response	Current path
Article reaction or comment	AI preference extraction → learned profile → future briefing scores
Research-thread direction shift	The next research session changes direction
Research-thread reaction or observation	Stored in thread history
Research-thread wrap-up	Thread closes with a conclusion
Response to an AI Observation	Stored in response memory; not automatically acted on yet

Tiered, purpose-fit model use, with real costs:

Stage	Model	Cost	Config-driven?
Daily ranking (production, single-stage)	grok-4-1-fast-reasoning	~\$0.314/run	Override ignored; intended fallback still used (see below)
Two-stage alternate (pre-filter → rank)	Claude Haiku → Sonnet	~\$0.001 pre-filter	Selectable per run; not the live path
Deep Dives	Claude Sonnet, on-demand	~\$0.25/session	No; hard-coded
AI Observations (daily)	Claude Haiku	~\$0.005-0.01/run	No; hard-coded

Stage	Model	Cost	Config-driven?
AI Observations (weekly)	Claude Sonnet, Sunday only	~\$0.07/week	No; hard-coded

These costs are current evaluation points, not fixed targets. Model choices are reviewed continuously for both cost and output quality. The intent is to move work to lower-cost models when they meet the required quality threshold, while leaving current choices in place long enough to evaluate them reliably.

Local capability is real, proven, and deliberately retained. The original architecture ran local-model calls in production. German's translation fallback still uses a live local provider as its final no-dependency option. Curator moved to cloud scoring after the per-run cost of a suitable cloud model became negligible. That was a convenience choice made possible by a quick backend swap, not a departure from the architecture.

One historical label remains from that change. The scoring script's free-tier mode is still called "ollama," although it currently uses keyword scoring rather than a live model. The label is imprecise; the behavior is intentional. The operational commitment remains that the system can return to local operation at zero marginal cost if cloud models become unavailable or too expensive. Re-verifying that swap end to end on the current EC2 environment is part of the planned model configuration work. It is a check on a proven capability, not a replacement for a missing one.

AI Observations is the platform's clearest working example of accumulating first and synthesizing later. Five observation types use model tiers matched to their cost and complexity. They work across a rolling 30-day baseline instead of reacting to a single day. The weekly higher-capability pass reads the accumulated history and asks, in effect, *what has been thought about this before?* No other domain can answer that question about itself yet.

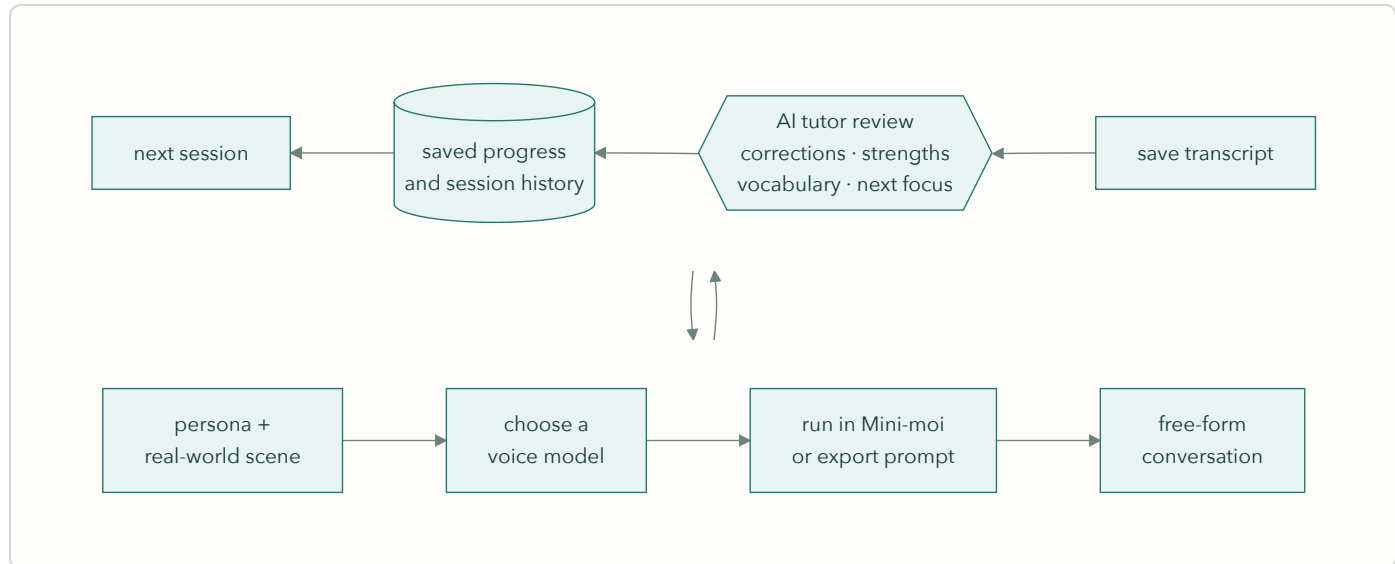
This is also where the thinking-record goal becomes concrete. Observation responses, Deep Dive motivations, and reactions are the raw material of that record. The capability shipped and is proven historically, but it is not currently scheduled on EC2. Nothing invokes it in the EC2 cron, the Docker configuration, or the CoS scheduler. Restoring the schedule is an operations follow-up under Principle 7.

Current model-override gap. Curator's `--model=` flag works for values it recognizes. Both production cron scripts currently pass an unrecognized value, so the override is ignored and the script uses its fallback default. That fallback still selects the intended production model, so there is no known impact on current Curator output. The gap is that the configured override cannot be relied on until the value is corrected. The defect is tracked and remains a clear example of Principle 7.

Multi-model challenge capability. Curator's Deep Dive pipeline supports one model drafting, other models cross-checking, and the first model reconciling the result. Live since June, it may overlap with Research Intelligence's Synthesizer+Challenger framework and Guild's `ChallengerService`. Direct confirmation is still needed. If they are the same underlying mechanism, the platform should give the capability one name and one home.

Known Deep Dive regression. Several generation scripts coexist while the frontend offers Scans, Deep Dive, and Deeper Dive as related but distinct options. At least four candidate scripts existed at the last count. Some may represent real duplication; others may support genuinely different features. The frontend grew faster than the backend was consolidated. The paths need to be traced before cleanup begins. This is regression refactoring, not a redesign.

German and Portuguese: one experience, converging implementations



German and Portuguese present the same core experience: the same navigation and interaction model over deliberately different first-stage implementations. Portuguese extended the language domain through a practical copy-and-adapt approach rather than a redesign. Its narrower initial scope emphasized the frontend, in-website voice, multi-user separation, and quick evaluation by real users. German retained the more setup-intensive automated review, Anki, lesson-plan, and mobile voice paths.

The two implementations are now converging around whichever approach proved stronger. Portuguese has a Postgres-backed translation cache, while German has a three-tier translation fallback (fast model → capable model → local model with no external dependency) and a stronger transcript-review tier. A third language will inherit the resulting shared pattern rather than either first implementation unchanged.

Voice conversation has delivered the greatest practical value in the language domains. Unscripted exchanges preserve the parts that scripted exercises remove: hesitation, getting stuck, using filler sounds, recovering, and continuing the interaction. This supports practice in conversational repair and recovery phrases, not only vocabulary and grammar. Real interactions can also be recreated afterward as scenarios, turning a difficult exchange into preparation for the next one.

Live persona conversation requires a model that understands the prompt and speaks the language fluently. That combination remains difficult, and quality from one provider can vary even within a single day, most plausibly under load. The reviewer model choice is therefore surfaced in the UI, with one default and other providers one click away. The learner is the first to notice a drop in fluency and is best placed to switch.

The current access paths involve a tradeoff. Exporting a persona prompt to a mobile model can provide the strongest voice experience, but copying, pasting, setup, and subscription requirements may confuse a new user. The in-browser path is easier to start but does not yet provide the same voice quality. Voice models, browser audio, and provider APIs are improving quickly, so this layer should remain adaptable and be revisited through the roadmap.

The transcript pipeline separates concerns differently. Deterministic parsing normalizes the transcript first. A strong model then analyzes it and produces feedback. On German's automated path, the analysis has concrete downstream effects: Anki card generation, the next lesson plan, and progress tracking.

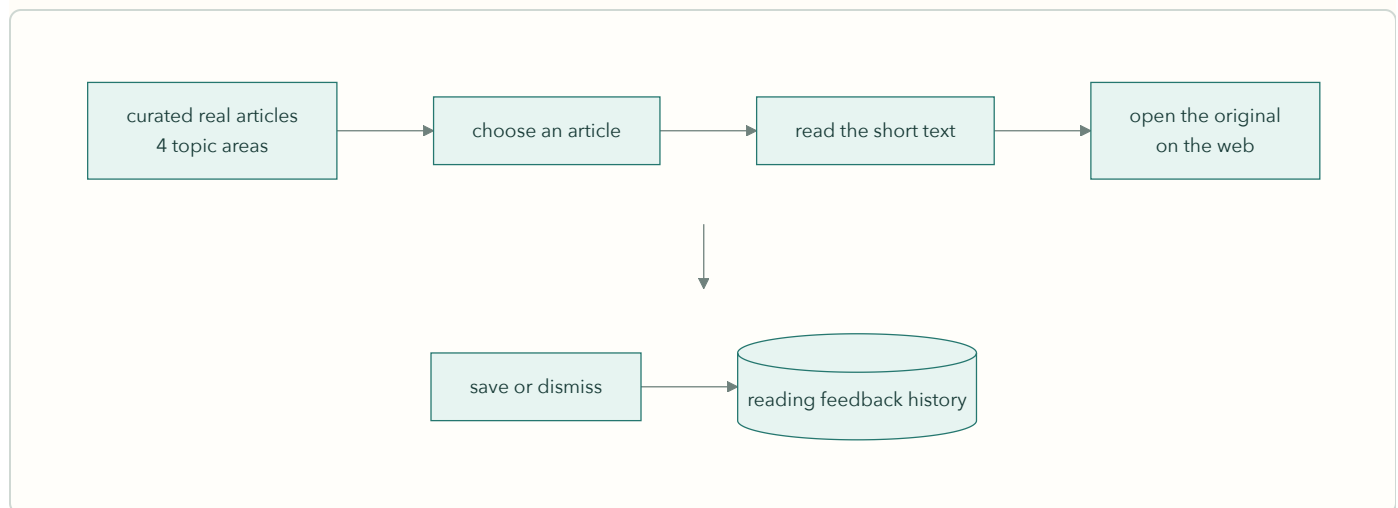
Model choice is split between the interface and the backend. Voice and review model choice are personal and visible at the point of use. Translation is not. The translation provider should be selected quietly on the backend through configuration. Both language domains still hard-code parts of that choice. Moving it to configuration is planned after the implementations converge.

Language reading should be engaging, varied, and contemporary. It combines language practice with cultural exposure rather than following a textbook sequence. Current material from everyday life, culture, news, and local sources provides a reason to read beyond the exercise itself. The goal is language plus culture. Contextual hints, translation, audio, and shorter excerpts are intended to preserve reading momentum when unfamiliar language would otherwise make the experience burdensome. They support authentic reading rather than replace it.

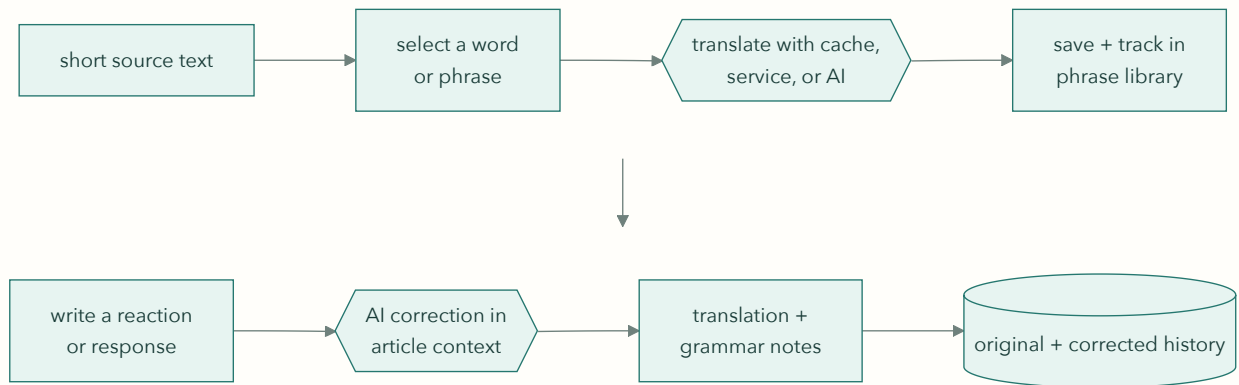
This requires freshness, variety, and source balance, but not Curator's full ranking apparatus. Curator is designed to make complex, nuanced, and sometimes conflicting information manageable across many sources. The language domains use a simpler source-variety cap because it better fits their purpose. Language learning is closer to learning an instrument than solving a decision problem; progress is felt through practice.

Writing is also anchored in current material so the learner has something worth responding to. Correction preserves both the original attempt and a more natural version, turning mistakes into material for later practice. Reading and writing therefore form a second, slower practice loop around current material:

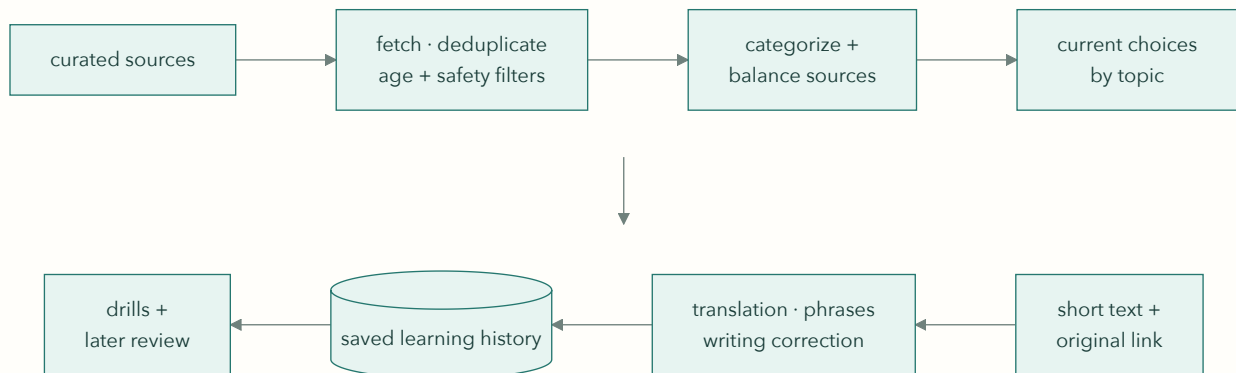
Reading and article feedback



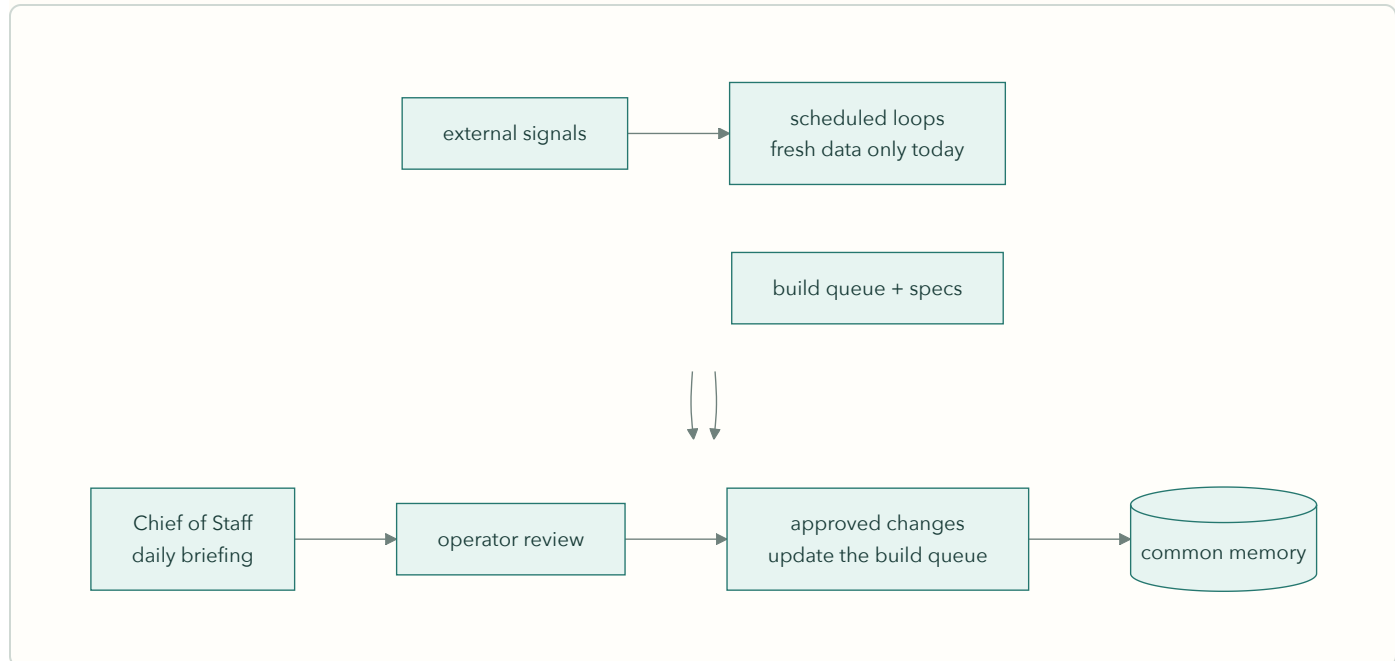
Vocabulary and writing



The material pipeline supports fresh, varied, and culturally relevant reading:



CoS and Guild: separate roles, active redesign



The common memory is not yet read by the scheduled loops. A revision returns to the briefing for another review; that return is stated here rather than drawn as a wide backward edge.

The separation clarified what each domain is for; neither was broken. The description below is current direction rather than settled architecture. Guild owns visibility into specifications, build work, quality, and operations. CoS owns cross-domain coordination and judgment. Keeping the planned Master Craftsman inside Guild preserves that role boundary. It provides local access to build context without expanding CoS access or mixing build coordination into its broader responsibilities.

Current CoS implementation. The chat backend and scheduled loops are ordinary Python code. No agent framework sits behind them yet. A bounded OpenClaw instance is planned as the Phase 1 agent layer. "Bounded" means restricted to mini-moi domains and the permission model below, not a general-purpose agent with unrestricted credentials. OpenClaw is intended to remain a swappable backend choice that is invisible to the user. Making that swap real is acknowledged integration work.

The first partner tasks are deliberately practical: operations escalations, ad hoc questions, and notes about areas of the application. The partner role should be earned through real daily work before it expands. CoS already has a common memory repository with episodic, semantic, and procedural tiers. That memory is independent of the agent placed above it, so changing the agent should not reset the accumulated record.

The partner contract. CoS follows a defined permission model: *propose freely, act within defined bounds, execute only with approval*. It may write to decision and action logs as part of normal operation, flag issues across domains, propose handoffs, retrieve documents, and file issues for identified defects. Writing to domain data, changing infrastructure, deploying, or executing a proposed handoff requires approval. Acting is bounded; informing is not.

The voice layer is designed for useful disagreement. The standing identity is short and already live in production: direct, curious before certain, comfortable saying "I don't know," and willing to disagree. It is not a cheerleader and not a search engine with manners. That identity applies to conversation and judgment tasks

such as chat, cross-domain health assessment, fit narratives, and novel suggestions. Mechanical operations and raw scoring remain accuracy-first. Partner is a mode, not a global override.

Current validation status. The founding identity document defines explicit success criteria, a two-to-three-week measurement period, and a subjective but decisive question: does the interaction feel like a working partner or a sophisticated tool? It also states the failure condition. If the result is indistinguishable from a well-prompted neutral agent, robotic agents are the better choice. The measurement has not run because the partner build is only starting. The design is ahead of the implementation, and this document treats it that way.

CoS is the first place planned bounded handoffs will be tested. The initial path is coordination with Guild's Master Craftsman agent. The operator should be able to ask CoS a question and let it gather information across agents instead of acting as the message bus personally. CoS is designed to inspect and communicate with any domain. It may instruct a domain only where that domain exposes an approved capability.

Four scheduled loops currently run: a twice-daily career scan and weekly-to-biweekly watches on language tooling, Curator's topic space, and competitive signals. Each loop evaluates only the external data fetched during that run. None reads the platform's accumulated memory before calling a model. Extending Curator's accumulate-and-synthesize pattern here is the next logical step because design queues, build status, and decisions need periodic reconciliation.

Learning Loop & Memory

Curator is the only fully proven instance of the learning loop today. Reactions update a learned profile, the dispatcher injects that profile into the next scoring run, and a weekly higher-capability pass reads 30 days of accumulated observations to identify what changed. The argument for the pattern is not only cost. Cheaper models with accumulated context can outperform more expensive models that lack it.

The idea predates the current memory work. The March 2026 production roadmap, written four months earlier, proposed a vector-indexed store so "every LLM call retrieves relevant personal context." It framed the work around the same question Curator's weekly pass asks today: *what has been thought about this before?* The current memory and intelligence roadmap is a more mature version of that idea, informed by production use.

CoS and Guild are the next places to apply the pattern. Their scheduled loops should read the decision log, action log, prior briefings, and build-queue history before calling a model, rather than scoring only fresh external data. A periodic higher-capability pass can then reconcile the record and ask what changed, what was decided but not completed, and what pattern no single day reveals. Design queues, build status, and decisions can degrade silently without that review. The language domains are the deliberate exception. Practice is not a decision problem, and adding this analytical machinery would work against their purpose.

Identity, Security & the Multi-Agent Working Model

Every domain sits behind a single portal. The portal owns session authentication and forwards identity to the appropriate backend. Domains never trust identity sent directly by the client. A July 2026 security review found and closed cases where a missing identity was treated as "match everything" rather than "match nothing." Details of the exposure remain outside this public document. The architectural lesson belongs here: nothing shipped without review, including the fix.

The multi-agent working model also separates roles. A decision authority approves every deployment. Other roles handle design and specification, implementation and deployment, memory and coordination, and independent review. The agent in any role may change over time. The roles are the durable structure.

Codex was introduced as a second coding agent alongside Claude Code. Both now perform build tasks within the same specification, review, and approval workflow. Adding a second coding agent changes who can perform implementation; it does not change the authority or review requirements attached to the role.

Open Design Questions

Stated plainly rather than resolved by default:

Does the platform depend on its current orchestration agent, or does the agent depend on the platform?

The March 2026 platform document states the latter. The current agent's self-description leans toward the former. This requires an explicit decision as bounded handoffs place the orchestration layer at the center of more workflows.

Is the multi-model challenge pattern in Curator's Deep Dive pipeline the same mechanism as Research Intelligence's Synthesizer+Challenger framework and Guild's `ChallengerService`? If so, one real capability deserves one name and one home.

Ongoing and Planned Follow-Up

- 1. Instrumented live session.** Run a real end-to-end session with monitoring and document which code paths execute, what they do, and which paths never fire. Static audits show what exists; this shows what is actually used. The session must explicitly verify local-model operation in Curator. Confirm the `--dry-run` local path and restore or verify the local-model fallback in production scoring. Fully local operation at zero marginal cost is a founding commitment, not an optional tier.
- 2. Confirm or rule out** whether Curator's Deep Dive pipeline, Research Intelligence's `generate_dive.py`, and Guild's `ChallengerService` are the same underlying capability.
- 3. Re-verify** the purpose of Curator's three coexisting Deep Dive-family scripts (Scans, Deep Dive, and Deeper Dive) before writing a cleanup spec.
- 4. Correct the Curator model override.** Production currently reaches the intended model through the fallback default, but the configured override is not recognized and should be corrected.
- 5. Correct `LLM_REGISTRY.md`.** The current registry is silent on CoS, Guild, and Portuguese; undercounts German; omits TTS; and overstates which model handles German's user-selectable review path.
- 6. Converge German and Portuguese.** Normalize both implementations in both directions before building French on the same template.
- 7. Move translation model choice into backend configuration.** Apply the change across both language domains after item 6.
- 8. Resolve the OpenClaw dependency direction.** Include the bounded agent-to-agent handoff design, starting with CoS and the Guild build agent.

9. **Build and measure the CoS partner layer.** The first work should remain operations escalations, ad hoc tasks, questions, and notes, with a bounded OpenClaw instance as the planned agent layer. Once the partner is in daily use, run the founding two-to-three-week measurement. Compare before and after, track novel suggestions, and make the explicit "partner or tool" judgment. That process turns the design from documented intent into a validated or rejected approach.
10. **Keep the per-domain flow diagrams current** as implementation changes. The durable user and operational flows belong here; detailed call graphs should remain separate so they can update on their own schedule.